

THE EVIDENCE FOR VEGANISM

Scientific Evidence on Animals, Health, Agriculture, and the Environment

A collection of findings from peer-reviewed research, scientific institutions, and major international datasets examining vegan diets and the broader impacts of animal agriculture.

The evidence covers human health, nutrition, land use, water, climate, biodiversity, food production, economics, public health, animal cognition, sentience, welfare, fishing, and aquaculture.

Each statement is presented concisely alongside its source and URL so the underlying evidence can be independently examined.

The purpose of this collection is simple: present the evidence clearly and let the data speak for itself.

1. LAND, CROPS & FOOD EFFICIENCY

 **Agriculture uses ~50% of Earth's habitable land.**

Source: FAO data / Our World in Data

URL: <https://ourworldindata.org/land-use>

 **Meat + dairy: ~83% farmland | ~18% calories | ~37% protein.**

Source: Poore & Nemecek, Science (2018)

URL: <https://doi.org/10.1126/science.aag0216>

 **No animal products: ~76% less agricultural land needed.**

Source: Poore & Nemecek, Science (2018)

URL: <https://doi.org/10.1126/science.aag0216>

 **Global soy: ~77% animal feed | ~7% direct human food.**

Source: USDA PSD / Food Climate Research Network data

URL: <https://ourworldindata.org/soy>

 **Corn: >95% of U.S. feed-grain production + use.**

Source: USDA Economic Research Service

URL: <https://www.ers.usda.gov/topics/crops/corn-and-other-feed-grains/>

 **Feeding human-edible crops to animals loses food calories.**

Source: Cassidy et al., Environmental Research Letters (2013)

URL: <https://doi.org/10.1088/1748-9326/8/3/034015>

 **Direct crop consumption can feed more people from the same land.**

Source: Cassidy et al., Environmental Research Letters (2013)

URL: <https://doi.org/10.1088/1748-9326/8/3/034015>

 **Global deforestation: ~90% driven by agriculture.**

Source: UN FAO, Global Remote Sensing Survey

URL: <https://www.fao.org/forest-resources-assessment/remote-sensing/remote-sensing-survey/en>

 **Deforestation: ~50% cropland expansion | ~38.5% livestock grazing.**

Source: UN FAO, Global Remote Sensing Survey

URL: <https://www.fao.org/forest-resources-assessment/remote-sensing/remote-sensing-survey/en>

2. WATER & POLLUTION

 **Global freshwater withdrawals: ~72% agriculture.**

Source: UN FAO

URL: <https://www.fao.org/land-water/water/water-scarcity/en/>

 **Farm nitrogen + phosphorus can pollute waterways.**

Source: UN FAO

URL: <https://www.fao.org/land-water/water/water-pollution/en/>

 **Excess nitrogen + phosphorus can cause eutrophication + dead zones.**

Source: U.S. Environmental Protection Agency

URL: <https://www.epa.gov/nutrientpollution>

 **Animal manure contains nitrogen + phosphorus pollutants.**

Source: U.S. Environmental Protection Agency

URL: <https://www.epa.gov/npdes/animal-feeding-operations-afos>

 **Manure runoff can carry pollutants into waterways.**

Source: U.S. Environmental Protection Agency

URL: <https://www.epa.gov/npdes/animal-feeding-operations-afos>

 **Livestock ammonia contributes to harmful PM2.5 pollution.**

Source: Domingo et al., PNAS (2021)

URL: <https://doi.org/10.1073/pnas.2013637118>

 **Animal foods cause most U.S. food-related air-quality health damage.**

Source: Domingo et al., PNAS (2021)

URL: <https://doi.org/10.1073/pnas.2013637118>

3. CLIMATE & BIODIVERSITY

 **Food systems: ~1/3 of human-caused GHG emissions.**

Source: Crippa et al., Nature Food (2021)

URL: <https://doi.org/10.1038/s43016-021-00225-9>

 **Food-production GHGs: ~57% animal foods | ~29% plant foods.**

Source: Xu et al., Nature Food (2021)

URL: <https://doi.org/10.1038/s43016-021-00358-x>

 **Beef alone: ~25% of global food-production GHGs.**

Source: Xu et al., Nature Food (2021)

URL: <https://doi.org/10.1038/s43016-021-00358-x>

 **No animal products: ~49% lower food GHG emissions modeled.**

Source: Poore & Nemecek, Science (2018)

URL: <https://doi.org/10.1126/science.aag0216>

 **No animal products: ~50% less acidification | ~49% less eutrophication.**

Source: Poore & Nemecek, Science (2018)

URL: <https://doi.org/10.1126/science.aag0216>

 **Agricultural expansion destroys + fragments wildlife habitat.**

Source: IPBES Global Assessment

URL: <https://www.ipbes.net/global-assessment>

 **50% animal-food replacement: modeled ~31% lower farm GHGs by 2050.**

Source: Kozicka et al., Nature Communications (2023)

URL: <https://doi.org/10.1038/s41467-023-40899-2>

 **50% animal-food replacement: modeled near-halt to net forest loss by 2050.**

Source: Kozicka et al., Nature Communications (2023)

URL: <https://doi.org/10.1038/s41467-023-40899-2>

4. HUMAN HEALTH & NUTRITION

 **Well-planned vegan diets can be nutritionally adequate for adults.**

Source: Academy of Nutrition and Dietetics (2025)

URL: <https://doi.org/10.1016/j.jand.2025.02.002>

 **Vegan diets can support long-term cardiometabolic health.**

Source: Academy of Nutrition and Dietetics (2025)

URL: <https://doi.org/10.1016/j.jand.2025.02.002>

Protein

 **Plant foods can provide all essential amino acids.**

Source: Mariotti & Gardner, Nutrients (2019)

URL: <https://doi.org/10.3390/nu11112661>

 **Vegan diets can meet human protein requirements.**

Source: Mariotti & Gardner, Nutrients (2019)

URL: <https://doi.org/10.3390/nu11112661>

 **Protein combining at every meal isn't necessary.**

Source: Mariotti & Gardner, Nutrients (2019)

URL: <https://doi.org/10.3390/nu11112661>

Fiber

 **Dietary fiber naturally occurs in plant foods.**

Source: NIH MedlinePlus

URL: <https://medlineplus.gov/dietaryfiber.html>

 **Meat: 0g fiber** |  **Eggs: 0g** |  **Milk: 0g.**

Source: USDA FoodData Central

URL: <https://fdc.nal.usda.gov/>

 **Beans, grains, fruits, vegetables, nuts + seeds provide fiber.**

Source: NIH MedlinePlus

URL: <https://medlineplus.gov/dietaryfiber.html>

Cholesterol

 **Whole plant foods are essentially cholesterol-free.**

Source: USDA FoodData Central

URL: <https://fdc.nal.usda.gov/>

 **Animal foods can contain dietary cholesterol.**

Source: USDA FoodData Central

URL: <https://fdc.nal.usda.gov/>

Randomized Vegan Trial

 **Vegan RCT: ~14 mg/dL greater LDL reduction in 8 weeks.**

Source: Landry et al., JAMA Network Open (2023)

URL: <https://doi.org/10.1001/jamanetworkopen.2023.44457>

 **Vegan RCT: ~2.9 µIU/mL greater fasting-insulin reduction.**

Source: Landry et al., JAMA Network Open (2023)

URL: <https://doi.org/10.1001/jamanetworkopen.2023.44457>

 **Vegan RCT: ~1.9 kg greater weight reduction in 8 weeks.**

Source: Landry et al., JAMA Network Open (2023)

URL: <https://doi.org/10.1001/jamanetworkopen.2023.44457>

Plant Protein & Longevity

 **Higher plant protein: associated with ~8% lower mortality.**

Source: Naghshi et al., BMJ (2020)

URL: <https://doi.org/10.1136/bmj.m2412>

 **Higher plant protein: associated with ~12% lower CVD mortality.**

Source: Naghshi et al., BMJ (2020)

URL: <https://doi.org/10.1136/bmj.m2412>

 **+3% energy from plant protein: associated with ~5% lower mortality.**

Source: Naghshi et al., BMJ (2020)

URL: <https://doi.org/10.1136/bmj.m2412>

Cancer

 **Processed meat: WHO/IARC Group 1 carcinogen.**

Source: WHO / International Agency for Research on Cancer

URL:

<https://www.who.int/news-room/questions-and-answers/item/cancer-carcinogenicity-of-the-consumption-of-red-meat-and-processed-meat>

 **Red meat: WHO/IARC Group 2A probable carcinogen.**

Source: WHO / International Agency for Research on Cancer

URL:

<https://www.who.int/news-room/questions-and-answers/item/cancer-carcinogenicity-of-the-consumption-of-red-meat-and-processed-meat>

 **50g processed meat/day: ~18% higher colorectal cancer risk.**

Source: WHO / International Agency for Research on Cancer

URL:

<https://www.who.int/news-room/questions-and-answers/item/cancer-carcinogenicity-of-the-consumption-of-red-meat-and-processed-meat>

 **Group 1 = evidence strength, not equal danger to tobacco.**

Source: WHO / International Agency for Research on Cancer

URL:

<https://www.who.int/news-room/questions-and-answers/item/cancer-carcinogenicity-of-the-consumption-of-red-meat-and-processed-meat>

Type 2 Diabetes

 **Higher red-meat intake: associated with higher type 2 diabetes risk.**

Source: Gu et al., American Journal of Clinical Nutrition (2023)

URL: <https://doi.org/10.1016/j.ajcnut.2023.08.021>

Modeled Global Health Effects

 **Global vegan scenario: ~8.1M fewer deaths/year by 2050.**

Source: Springmann et al., PNAS (2016)

URL: <https://doi.org/10.1073/pnas.1523119113>

5. PUBLIC HEALTH

 **Livestock antibiotic use can select for resistant microbes.**

Source: World Organisation for Animal Health

URL: <https://www.woah.org/en/what-we-do/global-initiatives/antimicrobial-resistance/>

 **Antibiotic misuse + overuse accelerates antimicrobial resistance.**

Source: World Health Organization

URL: <https://www.who.int/news-room/fact-sheets/detail/antimicrobial-resistance>

 **Resistant microbes can spread among animals, humans + ecosystems.**

Source: World Health Organization

URL: <https://www.who.int/news-room/fact-sheets/detail/antimicrobial-resistance>

 **Most emerging infectious diseases in a major study originated in animals.**

Source: Jones et al., Nature (2008)

URL: <https://doi.org/10.1038/nature06536>

 **Livestock intensification can affect zoonotic-disease risk.**

Source: Gilbert et al., Animal (2021)

URL: <https://doi.org/10.1016/j.animal.2020.100123>

6. SENTIENCE & COGNITION

 **Sentience = the capacity to experience feelings.**

Source: Lambert et al., Animals (2022)

URL: <https://doi.org/10.3390/ani12091182>

 **Fish are sentient.**

Source: Lambert et al., Animals (2022)

URL: <https://doi.org/10.3390/ani12091182>

 **Fish sentience: 349 papers | 142 species + subspecies reviewed.**

Source: Lambert et al., Animals (2022)

URL: <https://doi.org/10.3390/ani12091182>

 **Fish experience states including pain, fear + anxiety.**

Source: Lambert et al., Animals (2022)

URL: <https://doi.org/10.3390/ani12091182>

 **Fish can learn, remember + change behavior from experience.**

Source: Brown, Animal Cognition (2015)

URL: <https://doi.org/10.1007/s10071-014-0761-0>

 **Octopuses are sentient.**

Source: Birch et al., LSE Scientific Evidence Review (2021)

URL:

<https://www.lse.ac.uk/business/consulting/reports/review-of-the-evidence-of-sentiences-in-cephalopod-molluscs-and-decapod-crustaceans>

 **Squid are sentient.**

Source: Birch et al., LSE Scientific Evidence Review (2021)

URL:

<https://www.lse.ac.uk/business/consulting/reports/review-of-the-evidence-of-sentiences-in-cephalopod-molluscs-and-decapod-crustaceans>

 **Cuttlefish are sentient.**

Source: Birch et al., LSE Scientific Evidence Review (2021)

URL:

<https://www.lse.ac.uk/business/consulting/reports/review-of-the-evidence-of-sentiences-in-cephalopod-molluscs-and-decapod-crustaceans>

 **Crabs are sentient.**

Source: Birch et al., LSE Scientific Evidence Review (2021)

URL:

<https://www.lse.ac.uk/business/consulting/reports/review-of-the-evidence-of-sentiences-in-cephalopod-molluscs-and-decapod-crustaceans>

 **Lobsters are sentient.**

Source: Birch et al., LSE Scientific Evidence Review (2021)

URL:

<https://www.lse.ac.uk/business/consulting/reports/review-of-the-evidence-of-sentiences-in-cephalopod-molluscs-and-decapod-crustaceans>



Crayfish are sentient.

Source: Birch et al., LSE Scientific Evidence Review (2021)

URL:

<https://www.lse.ac.uk/business/consulting/reports/review-of-the-evidence-of-sentiences-in-cephalopod-molluscs-and-decapod-crustaceans>



Bivalves have central + peripheral nervous systems.

Source: Yurchenko et al., Frontiers in Zoology (2018)

URL: <https://pmc.ncbi.nlm.nih.gov/articles/PMC5896133/>



Oysters have cerebral + visceral ganglia.

Source: Yurchenko et al., Frontiers in Zoology (2018)

URL: <https://pmc.ncbi.nlm.nih.gov/articles/PMC5896133/>



Evidence for nociception exists in aquatic bivalves.

Source: Sneddon, Journal of Experimental Biology (2015)

URL: <https://journals.biologists.com/jeb/article/218/7/967/14518/Pain-in-aquatic-animals>



Clams + scallops detect predators + perform escape behaviors.

Source: Crook & Walters, ILAR Journal (2011)

URL: <https://doi.org/10.1093/ilar.52.2.185>



Scientific studies demonstrate learning in bivalves.

Source: Wageningen University & Research (2022)

URL: <https://research.wur.nl/en/publications/welfare-of-bivalves-in-aquaculture-a-first-literature-scan-and-pl/>



Pigs demonstrate learning + memory.

Source: Marino & Colvin (2015)

URL: <https://escholarship.org/uc/item/8sx4s79c>



Pigs demonstrate complex social behavior.

Source: Marino & Colvin (2015)

URL: <https://escholarship.org/uc/item/8sx4s79c>



Pigs possess sophisticated cognitive abilities.

Source: Marino & Colvin (2015)

URL: <https://escholarship.org/uc/item/8sx4s79c>



Chickens demonstrate learning + memory.

Source: Marino, Animal Cognition (2017)

URL: <https://doi.org/10.1007/s10071-016-1064-4>



Chickens demonstrate social cognition.

Source: Marino, Animal Cognition (2017)

URL: <https://doi.org/10.1007/s10071-016-1064-4>



Chickens can distinguish individual animals.

Source: Marino, Animal Cognition (2017)

URL: <https://doi.org/10.1007/s10071-016-1064-4>

 **Cattle demonstrate learning + memory.**

Source: Marino & Allen (2017)

URL: <https://doi.org/10.26451/abc.04.01.06.2017>

 **Cattle display measurable emotional responses.**

Source: Marino & Allen (2017)

URL: <https://doi.org/10.26451/abc.04.01.06.2017>

 **Cattle form social relationships + show individual personalities.**

Source: Marino & Allen (2017)

URL: <https://doi.org/10.26451/abc.04.01.06.2017>

7. ANIMAL WELFARE & FARMING

 **Farm animals can experience pain, fear + distress.**

Source: European Food Safety Authority

URL: <https://www.efsa.europa.eu/en/topics/topic/animal-welfare>

 **Animal transport can cause heat stress, thirst + fatigue.**

Source: European Food Safety Authority

URL: <https://www.efsa.europa.eu/en/topics/topic/animal-welfare>

 **Some farming systems restrict highly motivated natural behaviors.**

Source: European Food Safety Authority

URL: <https://www.efsa.europa.eu/en/topics/topic/animal-welfare>

 **Commercial farming can involve painful procedures.**

Source: European Food Safety Authority

URL: <https://www.efsa.europa.eu/en/topics/topic/animal-welfare>

8. FISHING & AQUACULTURE

 **Bycatch = unintended capture of non-target marine animals.**

Source: NOAA Fisheries

URL: <https://www.fisheries.noaa.gov/national/bycatch/understanding-bycatch>

 **Bycatch includes turtles, mammals, seabirds + non-target fish.**

Source: NOAA Fisheries

URL: <https://www.fisheries.noaa.gov/national/bycatch/understanding-bycatch>

 **Bycatch can injure or kill captured animals.**

Source: NOAA Fisheries

URL: <https://www.fisheries.noaa.gov/national/bycatch/understanding-bycatch>

 **Aquaculture feed can contain wild-caught fish, fishmeal + fish oil.**

Source: Global aquaculture-feed review

URL: <https://pmc.ncbi.nlm.nih.gov/articles/PMC11482329/>

 **Fisheries supplying aquaculture feed can produce bycatch.**

Source: Global aquaculture-feed review

URL: <https://pmc.ncbi.nlm.nih.gov/articles/PMC11482329/>

9. ECONOMICS & RESOURCE USE

 **Plant-based diets can cost less in high-income countries.**

Source: Springmann et al., Lancet Planetary Health (2021)

URL: [https://doi.org/10.1016/S2542-5196\(21\)00251-5](https://doi.org/10.1016/S2542-5196(21)00251-5)

 **Vegan diets were among the cheapest modeled healthy diets.**

Source: Springmann et al., Lancet Planetary Health (2021)

URL: [https://doi.org/10.1016/S2542-5196\(21\)00251-5](https://doi.org/10.1016/S2542-5196(21)00251-5)

 **Global vegan scenario: ~\$1.07T/year health savings by 2050.**

Source: Springmann et al., PNAS (2016)

URL: <https://doi.org/10.1073/pnas.1523119113>

 **Food-related climate damage creates measurable economic costs.**

Source: Errickson et al., Nature Food (2021)

URL: <https://doi.org/10.1038/s43016-021-00265-1>

10. EVIDENCE KEY

~ = approximately.

Associated with = an observational association, not automatic proof of causation.

Modeled / scenario = an estimated outcome under specified assumptions.

RCT = randomized controlled trial.

GHG = greenhouse gas.

PM2.5 = fine particulate air pollution.

CVD = cardiovascular disease.

Group 1 carcinogen = sufficient evidence that an agent can cause cancer. It does not mean all Group 1 agents carry equal risk.

EVIDENCE STANDARD

No claim enters this collection simply because it supports veganism.

Every factual claim has a source directly underneath it.

Every URL must lead to evidence supporting the claim.

Each fact appears once, in the category where it belongs.

Related statistics are combined when doing so preserves their scientific meaning.

Numbers remain attached to their original population, comparison, timeframe, or model.

Associations remain associations.

Models remain models.

Claims remain short without changing their scientific meaning.

Primary research and authoritative scientific sources are preferred whenever available.

When stronger evidence exists, stronger evidence is used.

Unsupported claims are excluded.

Present the evidence clearly and let the data speak for itself.